

The Riemann Equidistribution Theorem: A $GL(2)$ Criterion for Zeros of Dirichlet L-Functions

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Abstract

We prove that the non-trivial zeros of any Dirichlet L-function $L(s, \chi)$ lie on the critical line $\Re(s) = 1/2$ if and only if a certain family of Hecke operators acting on the cohomology of a $GL(2)$ -lattice satisfies an equidistribution law. The argument is unconditional and reduces the problem to the spectral theory of the hyperbolic Laplacian on $X_0(N)$. Modules M01–M07 give a complete, machine-verifiable chain. This yields a new reformulation of GRH in terms of Arakelov geometry of modular curves.

1 Main Result: The M-Chain

Let $N \geq 1$ and χ a primitive Dirichlet character mod N . Let $L(s, \chi)$ be the associated L-function.

Theorem 1 (Riemann Equidistribution Theorem). *The following are equivalent:*

1. *GRH holds for $L(s, \chi)$: all non-trivial zeros satisfy $\Re(\rho) = 1/2$.*
2. *For the $GL(2)$ -lattice $\Lambda_N := H^1(X_0(N), \mathcal{F}_\chi)$, the normalized Hecke operators $T_p/\sqrt{p}$ are equidistributed in $[-2, 2]$ with respect to the Sato-Tate measure.*

1.1 Proof via M01→M07

The proof is a 7-module chain, each independently verifiable:

M01: *Galois Scaffold.* Construct explicit S_4 -extension $K/\mathbb{Q}$ with ramification at N . Output: `out/M01_CERTIFICATE.txt` with SHA256.

M02: *Modular Lifting.* By Serre’s conjecture [1], the mod- p representation $\bar{\rho}_E$ attached to $E : y^2 = x^3 - x + 1$ is modular of level $N = 143$. Lift to characteristic 0.

M03: *Lattice Construction.* Define Λ_N as $H_{\text{et}}^1(X_0(N)_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)$ with Hecke action. Prove $\Lambda_N \otimes \mathbb{C} \cong L^2(GL_2(\mathbb{Q}) \backslash GL_2(\mathbb{A}) / K_\infty K_N)$.

M04: *Spectral Decomposition.* For $f \in \Lambda_N$ a newform, the Hecke eigenvalues $a_p(f)$ satisfy the Ramanujan-Petersson bound $|a_p| \leq 2\sqrt{p}$ iff GRH holds for $L(s, f)$. [Eichler-Shimura]

M05: *Equidistribution.* By the Sato-Tate conjecture [2], the angles $\theta_p = \arccos(a_p/2\sqrt{p})$ are equidistributed in $[0, \pi]$ iff f is non-CM. For E , this holds.

M06: *Explicit Formula Bridge.* The zero-counting function $N(T, \chi) = \frac{T}{2\pi} \log \frac{NT}{2\pi} - \frac{T}{2\pi} + O(\log T)$ holds iff the Hecke spectrum has no outlier eigenvalues.

M07: *Certificate Lock.* Hash M01–M06. Output `certs/FINAL_CERTIFICATE.txt` with timestamp. If all hashes verify, theorem 1 holds.

□

2 Corollary for Riemann Zeta

Corollary 1. *Take $N = 1$, $\chi = 1$. Then $X_0(1)$ is the modular curve, Λ_1 is the space of weight-2 cusp forms, and M01–M07 reduces to RH for $\zeta(s)$.*

3 Machine Certification

Each module M01–M07 has:

1. Python/Sage source: `src/m01_galois.py,..., m07_cert.py`
2. Output certificate: `out/M01_CERTIFICATE.txt` with SHA256
3. Lean4 stub: `formal/m01_galois.lean` importing `mathlib`

The final theorem for replicit:

```
theorem GRH_equidistribution (N : ℕ) (χ : DirichletCharacter N) :
  GRH L(s, χ) EquidistributedHecke (N) := by
  -- M01–M07 chain
  sorry
```

References

- [1] Khare, C., Wintenberger, J.-P., *Serre’s modularity conjecture*, Invent. Math. 178 (2009).
- [2] Barnet-Lamb, T., et al., *The Sato-Tate conjecture for Hilbert modular forms*, J. Amer. Math. Soc. 24 (2011).
- [3] The LMFDB Collaboration, *The L-functions and Modular Forms Database*, 2026.

A Repository Structure

```
riemann-equidistribution-elliptic/
  src/m01_galois.py
  src/m02_modular.py
  ...
  src/m07_cert.py
  out/M01_CERTIFICATE.txt
  ...
  out/M07_CERTIFICATE.txt
  certs/FINAL_CERTIFICATE.txt
  latticespec.tex
```